

Path Planning Report

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Overview

I started by installing matplotlib. The project needs it to draw the cones, the car and the path, and the first run failed with a missing-module error until I installed it through pip.

The original zip came with a placeholder planner that just drew a straight line ahead and ignored every cone. It also had a scenario file with 20 test cases, a simple viewer and a small runner script.

Part 1

I replaced the placeholder with a cone-based planner built around one rule: travelling along the path, blue is always on the left and yellow is always on the right. The planner pairs blue and yellow cones, closest pairs first, and treats each pair as a gate. The midpoint of a gate becomes a waypoint, and the travel direction there is the gate turned ninety degrees. A cone with no partner becomes a waypoint shifted half a track width toward the inside of the track, and when no pair is visible it assumes a two-metre-wide track. The waypoints are sorted by distance from the car, and any that would make the path double back are dropped. A smooth cubic spline then runs through them and continues straight on. The result is resampled every quarter of a metre up to eight metres, and any point closer than forty centimetres to a cone is pushed away from it.

Part 2

For the second part, three cones on one side, no special case was needed. When one side has three cones, the middle cone takes its direction from the cones before and after it, so the path can follow a curve. I then added test scenarios. My first batch of random cases turned out to be physically impossible, with cones on the wrong sides of the car, so I replaced them with hand-placed ones. The final set is scenarios 21 to 35. Each has exactly three cones on one side, and together they cover straights and left and right curves, cones on the inside and the outside of a turn, cases with only one colour visible, and cases with an extra cone ahead that has no partner.

Visulising

Finally, view_all is a small script that draws every scenario in one window. Each box shows the cones, the car, the old straight-line placeholder as a dashed line and the new path in green.

Problems and Solutions

The early versions failed in several kinds of cases, and each needed its own fix. In scenarios 2 and 3 the path went up and to the right and never turned right, so I paired blue and yellow cones into gates and made the path pass through the middle of each gate at a right angle to it. In scenarios 4,

9, 12, 17 and 20 there was only one cone of one colour, so the path never changed direction; I shifted that cone half a track width toward the inside of the track and used it as a waypoint. In scenarios 5, 8, 13 and 16 only two cones of the same colour were visible, so the planner put yellow on the left in 5, blue on the right in 8 and made a U-turn between the cones in 13. I fixed this by always keeping blue on the left and yellow on the right, taking the direction from the line through the two cones, and dropping waypoints that make the path double back. In scenario 14 the planner found no path because it was limited to 75 degrees, so I built the path from the cones and used the car's heading only when no cones gave a direction. In case 15 a cone behind a gate still created a waypoint, which made the path hook backwards, so I used the previous cone to set the gate's exit direction instead.

Algorithms

The planner follows one rule: blue is always on the left and yellow on the right.

- **Pairing:** I pair blue and yellow cones, closest first, and use each pair's midpoint as a waypoint.
- **Lone cones:** a cone with no partner is shifted half a track width toward the inside of the track. I assume a two-meter track when no pair is visible.
- **Filtering:** I sort the waypoints by distance from the car and drop any that would make the path double back.
- **Spline:** I run a cubic Hermite spline through the waypoints and continue straight past the last one.
- **Resampling:** I resample the path every quarter of a meter up to eight meters.
- **Clearance:** I push any point closer than forty centimeters to a cone away from it.